#**Hands-On Activity 10 | Explaining Visually to Influence with Data**

Name : Andrew S. Doloso Jr.
Course Code and Title : CPE031-Visualizations and Data Analysis
Date Submitted : 10-10-26
Instructor : Engr. Jimlord Quejado

---

1. Objectives:

This activity aims to help students demonstrate how to influence and communicate effectively with data by identifying audience needs, crafting a clear message, and telling a story visually through charts.

2. Intended Learning Outcomes (ILOs):

By the end of this activity, students should be able to:

1. Analyze the audience to understand their level of data knowledge and needs.
2. Evaluate how well a message and visualization communicate insights.
3. Create a clear and meaningful data story using Python visualization tools.

3. Discussions:

Data storytelling goes beyond showing charts it's about influencing decisions and sparking understanding.

Know Your Audience: Decision-makers may prefer clear summaries, while analysts like detailed data.

Know Your Message: Focus on the why it matters part of the story.

Tell a Story: Use preattentive attributes color, size, and position to guide attention.

Structure Your Story: Like Aristotle's rhetorical triangle, combine logic (logos), emotion (pathos), and credibility (ethos).

4. Procedures:

Step 1. Import Libraries and Create Sample Data

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Sample sales dataset
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [200, 250, 180, 300, 280, 350],
    "Marketing_Spend": [50, 60, 45, 70, 65, 80]
```

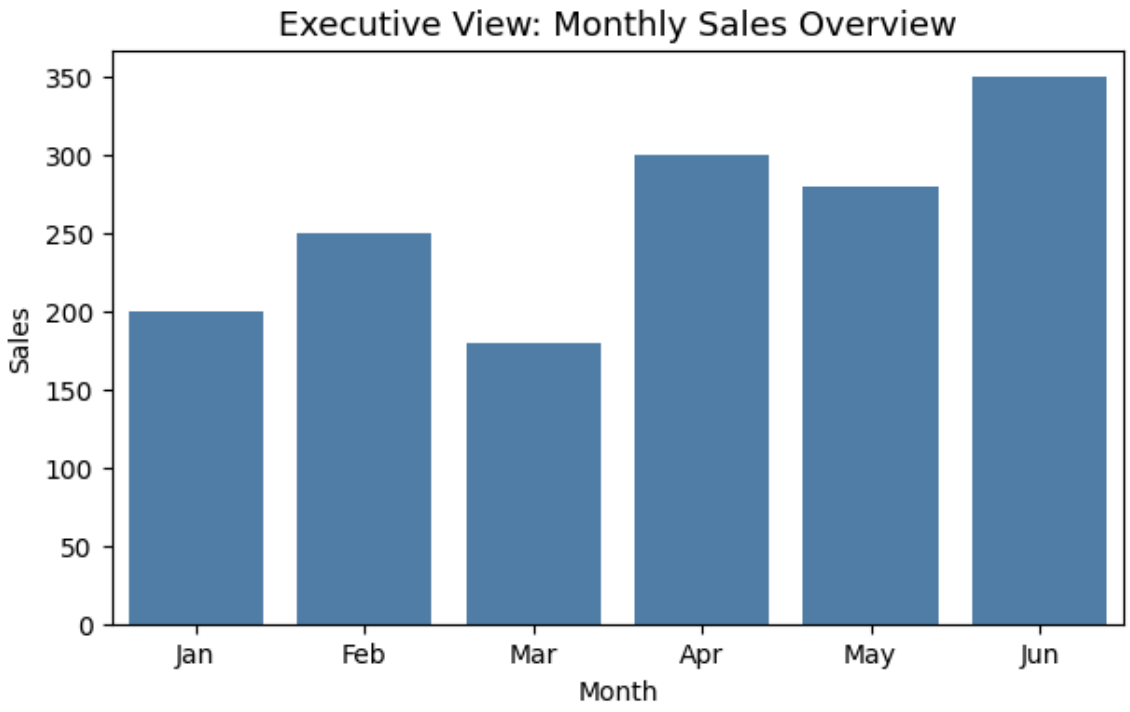
```
}
df = pd.DataFrame(data)
df
```

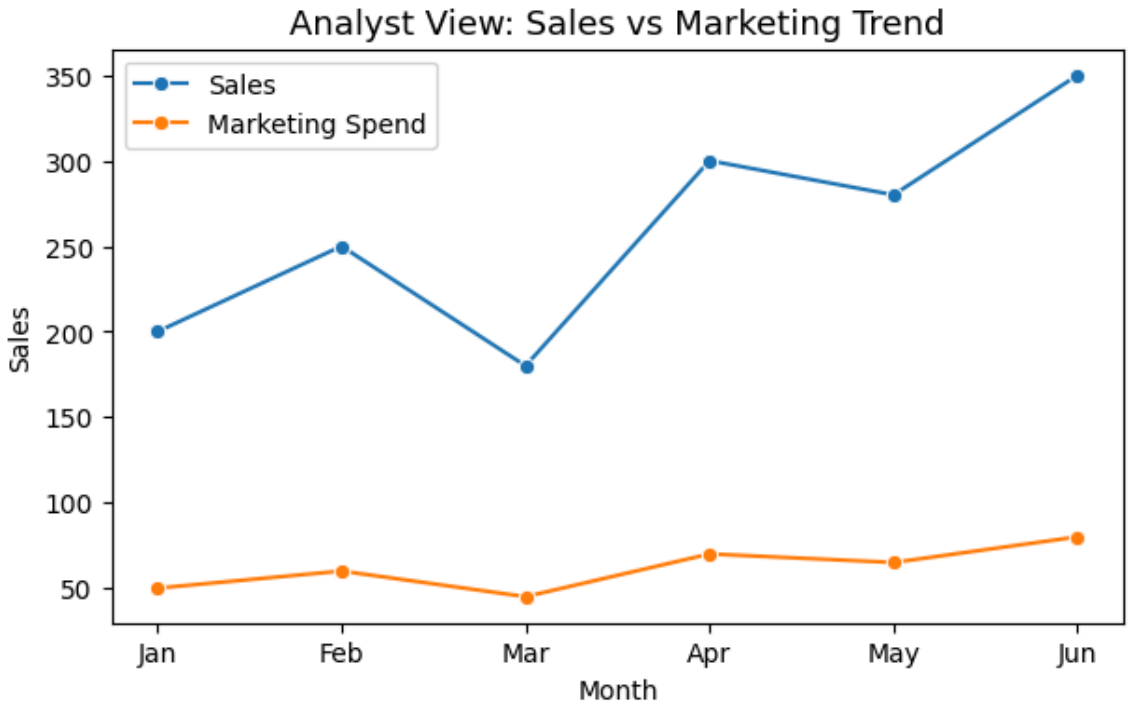
	Month	Sales	Marketing_Spend
0	Jan	200	50
1	Feb	250	60
2	Mar	180	45
3	Apr	300	70
4	May	280	65
5	Jun	350	80

Step 2. Know Your Audience — Compare Chart Styles

```
# Executive audience: prefer clarity (simple bar)
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="steelblue")
plt.title("Executive View: Monthly Sales Overview", fontsize=13)
plt.show()

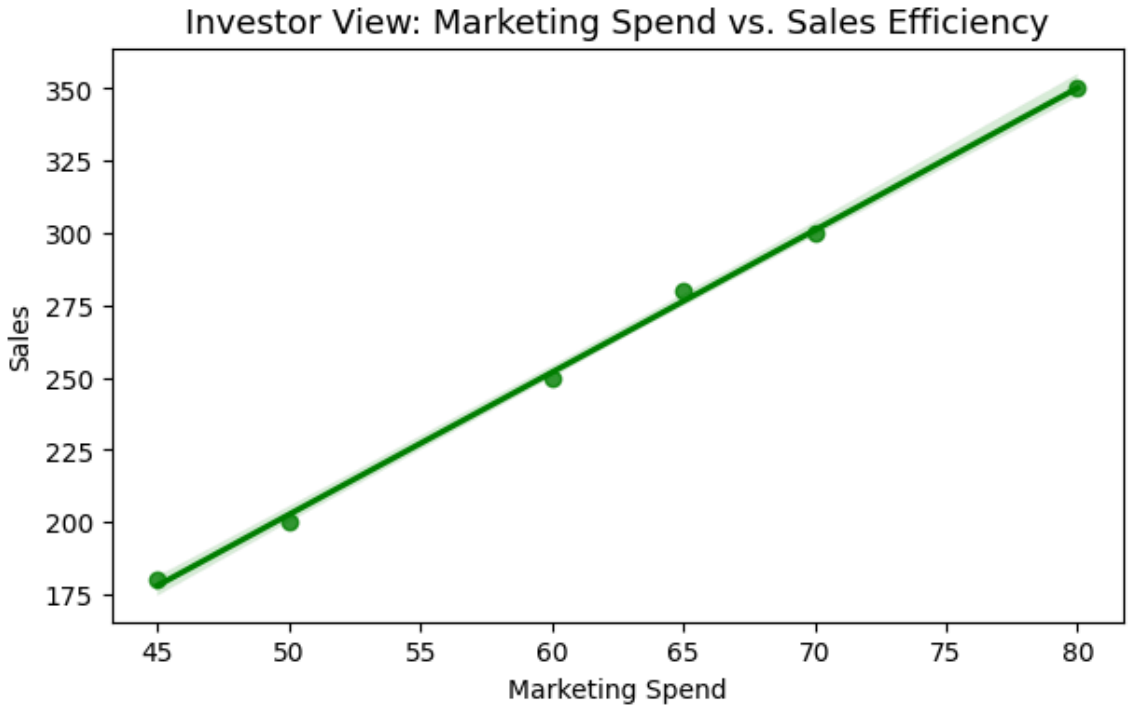
# Analyst audience: prefer detailed trends (line chart)
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o",
         label="Marketing Spend")
plt.title("Analyst View: Sales vs Marketing Trend", fontsize=13)
plt.legend()
plt.show()
```





Task 1: Create one additional visualization suited for a different audience (e.g., a customer, investor, or teammate).

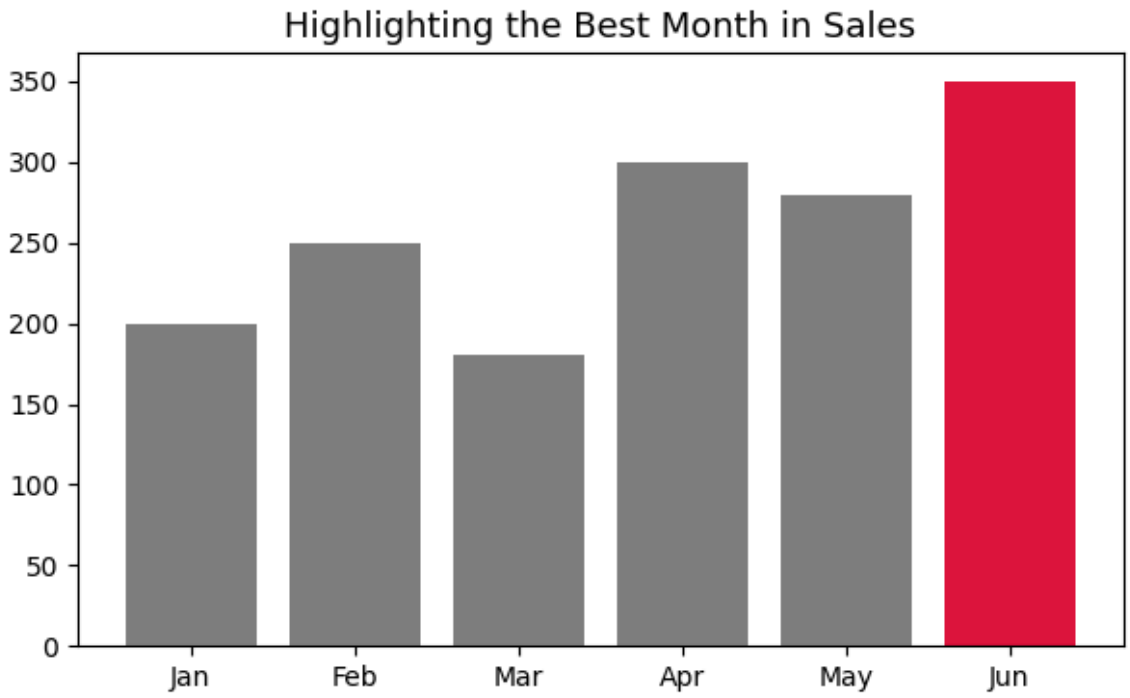
```
#Code Here
plt.figure(figsize=(7, 4))
sns.regplot(x="Marketing_Spend", y="Sales", data=df, color="green",
         marker="o")
plt.title("Investor View: Marketing Spend vs. Sales Efficiency", fontsize=13)
plt.xlabel("Marketing Spend")
plt.ylabel("Sales")
plt.show()
```



Step 3. Know Your Message — Highlight What Matters

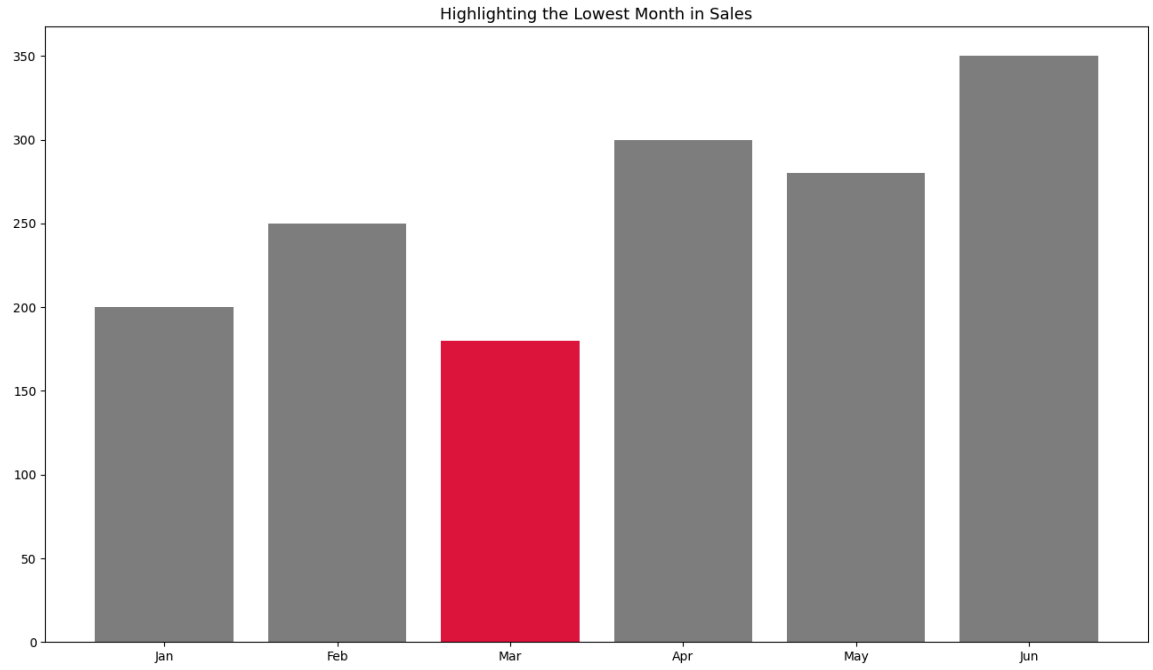
```
# Highlight the highest sales month (preattentive color cue)
highlight = df["Sales"].idxmax()

colors = ["gray" if i != highlight else "crimson" for i in range(len(df))]
plt.figure(figsize=(7,4))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Highlighting the Best Month in Sales", fontsize=13)
plt.show()
```



Task 2: Use color or annotations to emphasize a key insight (e.g., lowest point, sudden increase).

```
#Code Here
highlight = df["Sales"].idxmin()
colors = ["gray" if i != highlight else "crimson" for i in range(len(df))]
plt.figure(figsize=(16,9))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Highlighting the Lowest Month in Sales", fontsize=13)
plt.show()
```

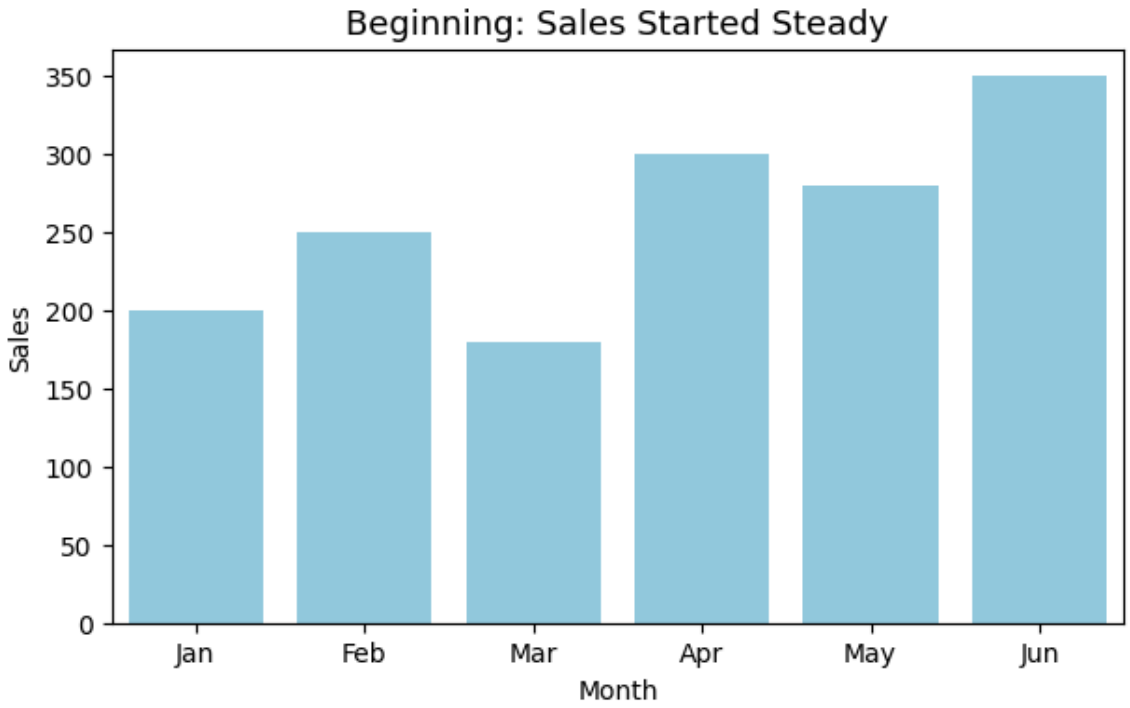


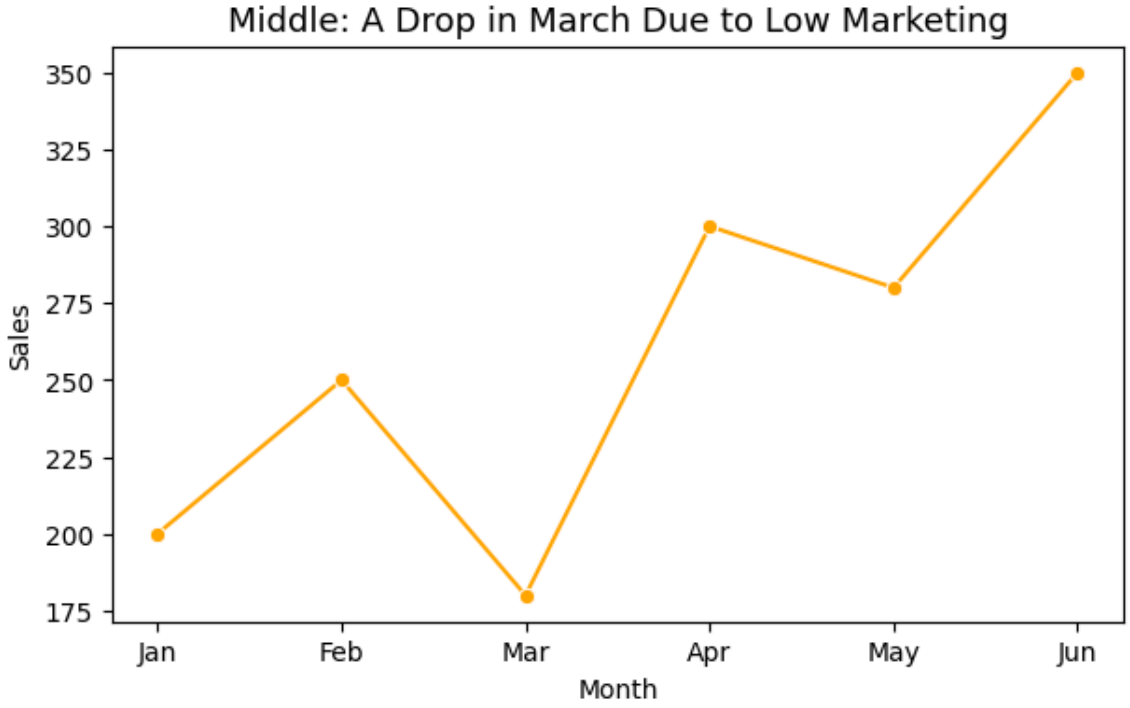
Step 4. Storytelling with Charts — Beginning, Middle, End

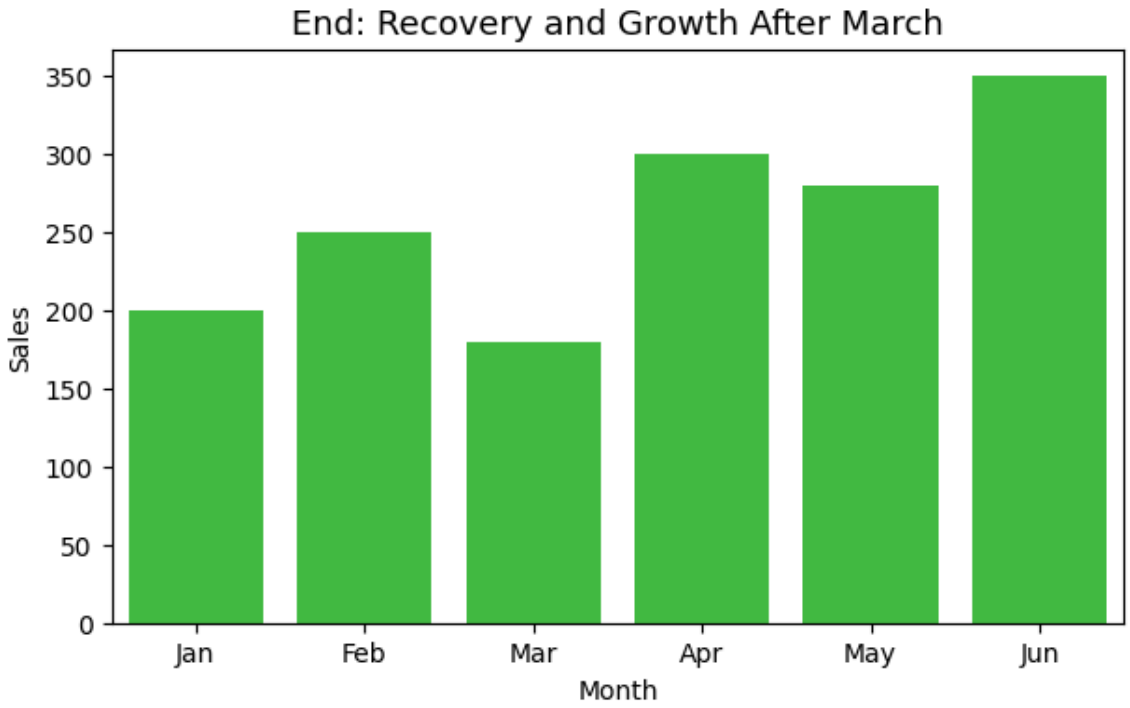
```
# Beginning: Context
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="skyblue")
plt.title("Beginning: Sales Started Steady", fontsize=13)
plt.show()

# Middle: Change or Challenge
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, color="orange", marker="o")
plt.title("Middle: A Drop in March Due to Low Marketing", fontsize=13)
plt.show()

# End: Resolution
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="limegreen")
plt.title("End: Recovery and Growth After March", fontsize=13)
plt.show()
```


Beginning: Sales Started Steady
Sales
Month
Jan
Feb
Mar
Apr
May
Jun
0
50
100
150
200
250
300
350


Middle: A Drop in March Due to Low Marketing
Sales
Month
Jan
Feb
Mar
Apr
May
Jun
175
200
225
250
275
300
325
350



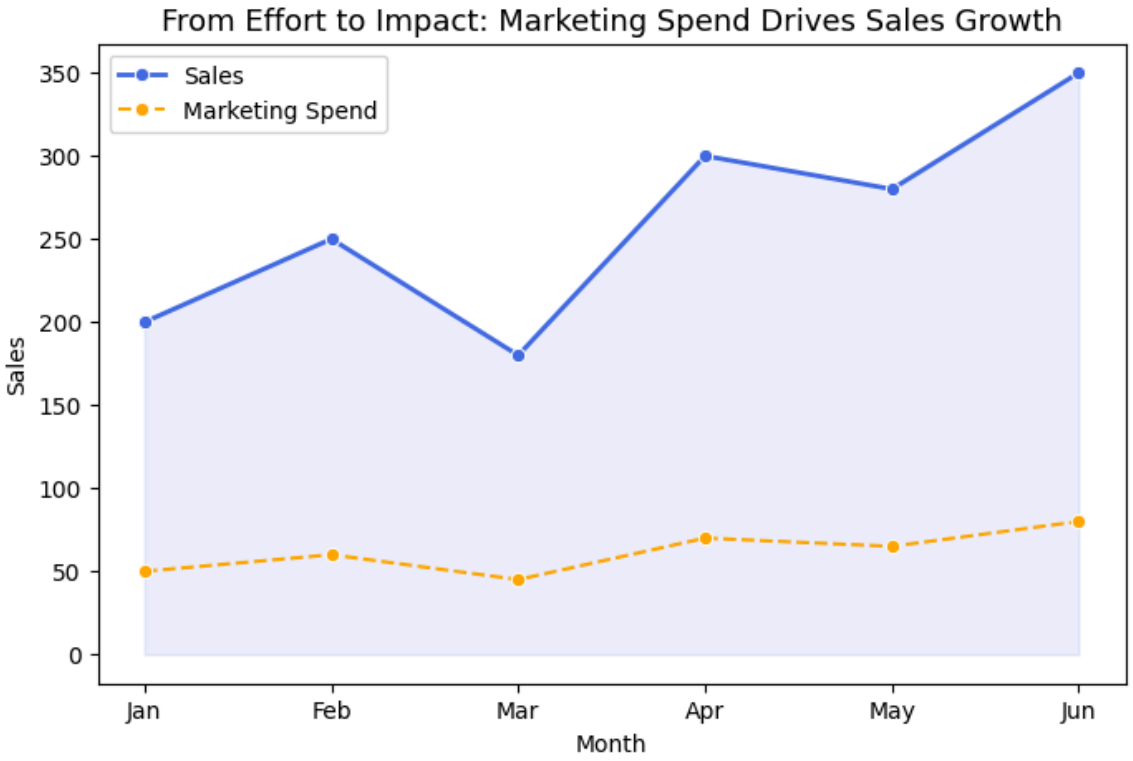
Task 3: Write one or two lines describing your data story from beginning → middle → end.

Sales were steady in January and increased in February, but dropped in March. They went up again in April, dropped a little in May, and reached their highest peak point in June.

Step 5. Bringing It All Together — Presentation Design

```
# Simple combined story chart
plt.figure(figsize=(8,5))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", linewidth=2,
         color="royalblue", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o",
         linestyle="--", color="orange", label="Marketing Spend")

plt.fill_between(df["Month"], df["Sales"], alpha=0.1, color="royalblue")
plt.title("From Effort to Impact: Marketing Spend Drives Sales Growth",
         fontsize=13)
plt.legend()
plt.show()
```



---

5. Supplementary Activity:

1. Use a small dataset available online (kaggle, google dataset, etc.).
2. Visualize it using a bar chart.
3. Highlight one important data point (e.g., the highest or lowest value) using a distinct color.
4. Add a title and annotation that tell the story behind the data.

```
#Code here
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# 1. Load the Southeast Asian Population dataset
data = {'Country': ['Indonesia', 'Philippines', 'Vietnam','Thailand',
        'Malaysia'],'Population': [281.2, 114.9, 100.4, 71.8, 35.1]
}

df = pd.DataFrame(data)

# population by country
country_population = df.groupby('Country')['Population'].sum().reset_index()

# 3. Highlight the highest population
highlight_idx = country_population['Population'].idxmax()

colors = ['gray' if i != highlight_idx else 'lightgreen'
          for i in range(len(country_population))]

# 2. bar chart
```

```
plt.figure(figsize=(10, 5))
plt.bar(country_population['Country'],country_population['Population'],color=colors)

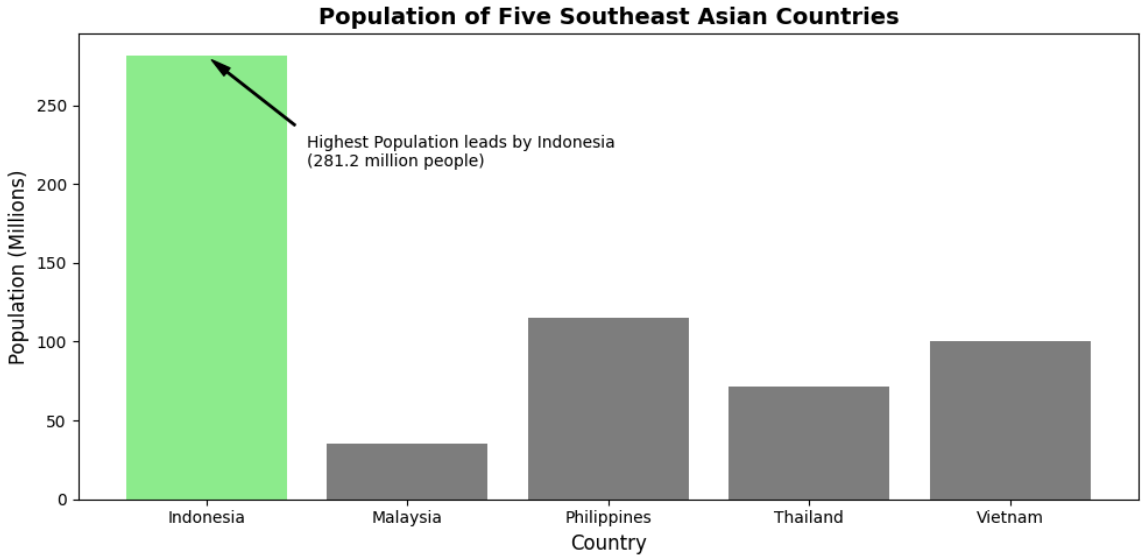
# 4. title and annotation that tell the story
plt.title('Population of Five Southeast Asian Countries',
          fontsize=14,weight='bold')

plt.xlabel('Country', fontsize=12)
plt.ylabel('Population (Millions)', fontsize=12)

max_country = country_population.loc[highlight_idx, 'Country']
max_val = country_population.loc[highlight_idx, 'Population']

plt.annotate(f'Highest Population leads by {max_country}\n'f'({max_val:.1f}
         million people)',
    xy=(highlight_idx, max_val), xytext=(highlight_idx + 0.5, max_val - 70),
    arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6),
    fontsize=10,backgroundcolor='white')

plt.tight_layout()
plt.show()
```



6. Conclusion/Learnings/Analysis:

In this HOA activity, I learned that data visualization is not just about creating charts but also about presenting information clearly and meaningfully. I learned that choosing the right colors, titles, and annotations helps the audience understand the important data. This activity helped me how to use Python to create charts that communicate insights and help people make better decisions.